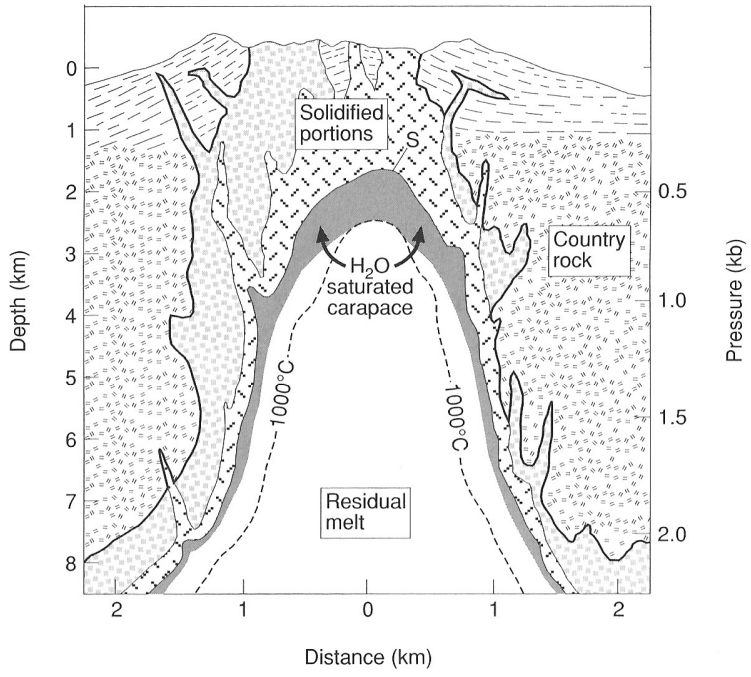
The origin of hydrothermal fluids from magma (magmatic-hydrothermal)

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All melts contain small volumes of volatiles such as water, which are dissolved in the melt. This water is exsolved in the last stages of crystallization of igneous bodies (e.g. a granite intrusion), forming magmatic-hydrothermal aqueous fluids. These can carry incompatible elements (such as U, Th, Sn) from the melt, and equally dissolve additional elements from the country rock (due to the temperature of fluids), thus forming a concentrated aqueous solution capable of carrying and eventually depositing these elements.



Robb (2005)

What happens to the volatiles, mostly water, as the magma crystallizes? Imagine that you have 100 tonnes of magma and of that, 4% (4 tonnes) is volatiles and 96% is silicates. The volatiles are dissolved in the melt and do not crystallize out. As the silicates crystallize the volatiles are left in the remaining melt. But the melt can only contain up to 10% volatiles dissolved in it; any more and the volatiles separate out as a separate phase, called free water. The table below provides a simple calculation of this process.

Complete the table on the next page. What percentage of magma has crystallized before any free water is formed?

Solution:

Melt crystallized (tonnes)	Remaining melt (tonnes)	Volatiles (tonnes)	Silicate melt (tonnes)	Volatiles (%)	Free water (tonnes)
0	100	4	96	4/100 = 4%	0
10	90	4	86	4/90 = 4.4%	0
20	80	4	76	4/80 = 5%	0
30	70	4	66	4/70 = 5.7%	0
40	60	4	56	4/60 = 6.6%	0
50	50	4	46	4/50 = 8 %	0
60	40	4	36	4/40 = 10%	0
70	30	3*	26	3/30#	1
80	20	2	16	2/20	2
90	10	1	6	1/10	3
100	0	0	0	0/0	4

Notes:
*: at this stage, the melt can hold a maximum of 10% water (at saturation), hence 10% of 30/20/10/0 tonnes
#: saturation is reached, and the % of volatiles does not change any more; any excess is exsolved